

Week 1 Assignment Report

Java Fundamentals

Name: Mounika

College: Narayana Engineering College, Gudur

Branch: Computer Science and Engineering

Internship: Java Internship **Submitted**

By: Mounika

Objective

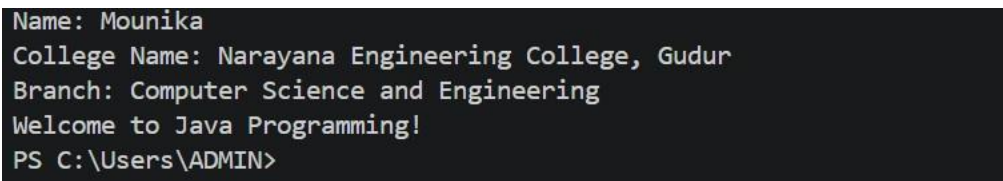
The main objective of this assignment is to understand the fundamental concepts of Java programming and develop basic programming skills. This assignment focuses on learning Java syntax, variables, data types, user input using the Scanner class, arithmetic operators, and type casting. It also helps in understanding how to write, compile, and execute Java programs in a proper development environment. By completing these tasks, students gain practical experience in writing simple Java programs, improve logical thinking and problem-solving skills, and build a strong foundation for advanced Java programming concepts that will be covered in future assignments.

Task 1: java setup & Hello World

Code:

```
public class Task1 {    public static void  
main(String[] args) {  
  
    System.out.println("Name: Mounika");  
  
    System.out.println("College Name: Narayana Engineering College, Gudur");  
  
    System.out.println("Branch: Computer Science and Engineering");  
  
    System.out.println("Welcome to Java Programming!");  
  
    }  
}
```

Output:



```
Name: Mounika  
College Name: Narayana Engineering College, Gudur  
Branch: Computer Science and Engineering  
Welcome to Java Programming!  
PS C:\Users\ADMIN>
```

Explanation:

This program is the first Java program that demonstrates the basic structure of a Java application. It displays the student's name, college name, branch, and a welcome message using `System.out.println()`. It helps in understanding the basic syntax and output statements in Java.

Task 2 : Variable & Data Types

Code:

```
public class Task2 {    public static void  
main(String[] args) {  
  
    // Variables  
  
    String name = "Mounika";  
int age = 20;    float height =  
5.3f;    double weight =  
52.5;    double cgpa = 9.33;  
char gender = 'F';    boolean  
isStudent = true;  
  
    // Display Output  
  
    System.out.println("Name: " + name);  
    System.out.println("Age: " + age);  
    System.out.println("Height: " + height + " feet");  
    System.out.println("Weight: " + weight + " kg");  
    System.out.println("CGPA: " + cgpa);  
    System.out.println("Gender: " + gender);  
    System.out.println("Is Student: " + isStudent);  
}  
}
```

Output:

```
Name: Mounika  
Age: 20  
Height: 5.3 feet  
Weight: 52.5 kg  
CGPA: 9.33  
Gender: F  
Is Student: true
```

Explanation:

This program demonstrates the use of Java variables and different data types such as `String`, `int`, `float`, `double`, `char`, and `boolean`. Each variable stores a specific type of information and the program displays all the stored values in a neat format. This helps in understanding how data is stored and used in Java.

Task 3 : Scanner class

Code:

```
import java.util.Scanner; public class
Task3 {    public static void
main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter Full Name: ");

    String name = sc.nextLine();

    System.out.print("Enter Age: ");    int
age = sc.nextInt();    sc.nextLine();

    System.out.print("Enter College Name: ");

    String college = sc.nextLine();

    System.out.print("Enter Branch: ");

    String branch = sc.nextLine();

    System.out.print("Enter City: ");

    String city = sc.nextLine();

    System.out.println("\n----- Student Details -----");

    System.out.println("Full Name : " + name);

    System.out.println("Age      : " + age);

    System.out.println("College   : " + college);

    System.out.println("Branch    : " + branch);

    System.out.println("City      : " + city);
```

```
        sc.close();
    }
}
```

Output:

```
Enter Full Name: N.Mounika
Enter Age: 20
Enter College Name: Narayana Engineering College, Gudur
Enter Branch: Computer Science And Engineering
Enter City: Nellore

----- Student Details -----
Full Name : N.Mounika
Age       : 20
College   : Narayana Engineering College, Gudur
Branch    : Computer Science And Engineering
City      : Nellore
```

Explanation:

This program uses the `Scanner` class to accept input from the user through the keyboard. It collects details such as name, age, college, branch, and city, and then displays them in a formatted manner. This helps in learning user input handling and output formatting in Java.

Task 4 : Operators Practice

Code:

```
import java.util.Scanner; public class
Task4 {    public static void
main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter First Number: ");    int
    num1 = sc.nextInt();

    System.out.print("Enter Second Number: ");

    int num2 = sc.nextInt();
```

```
System.out.println("\n----- Calculator Results -----");

System.out.println("Addition = " + (num1 + num2));

System.out.println("Subtraction = " + (num1 - num2));

System.out.println("Multiplication = " + (num1 * num2));

System.out.println("Division = " + (num1 / num2));

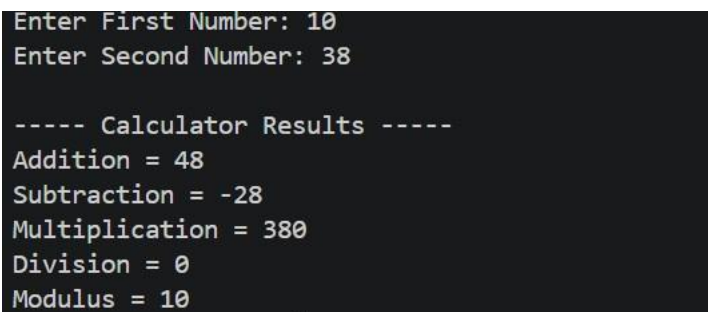
System.out.println("Modulus = " + (num1 % num2));


    sc.close();

}

}
```

Output :



```
Enter First Number: 10
Enter Second Number: 38

----- Calculator Results -----
Addition = 48
Subtraction = -28
Multiplication = 380
Division = 0
Modulus = 10
```

Explanation:

This program performs basic arithmetic operations using two numbers entered by the user. It calculates addition, subtraction, multiplication, division, and modulus using Java operators. This task helps in understanding arithmetic operators and mathematical calculations in Java.

Task 5 : Type Casting

Code:

```
public class Task5 {    public static void
main(String[] args) {    // Implicit
Type Casting    int number = 25;

    double doubleNumber = number;

    System.out.println("Implicit Type Casting");

    System.out.println("Before Casting (int): " + number);

    System.out.println("After Casting (double): " + doubleNumber);

    System.out.println();

// Explicit Type Casting
double marks = 95.75;    int
intMarks = (int) marks;

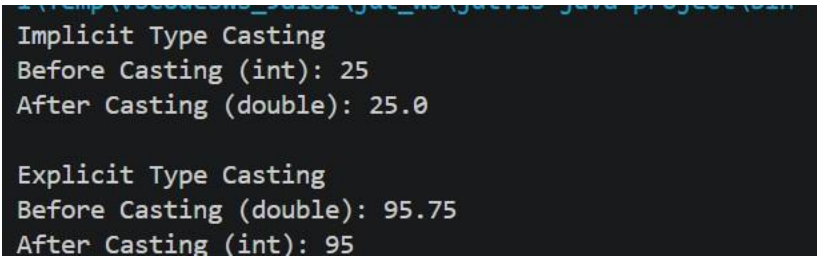
    System.out.println("Explicit Type Casting");

    System.out.println("Before Casting (double): " + marks);

    System.out.println("After Casting (int): " + intMarks);

}
}
```

Output:



```
Implicit Type Casting
Before Casting (int): 25
After Casting (double): 25.0

Explicit Type Casting
Before Casting (double): 95.75
After Casting (int): 95
```

Explanation:

This program demonstrates both implicit and explicit type casting in Java. Implicit type casting automatically converts a smaller data type into a larger one, while explicit type casting manually converts a larger data type into a smaller one. This task helps in understanding data type conversion and how it affects values during execution.

Task 6 : Java Fundamentals Practices

Code:

```
import java.util.Scanner; public class
Task6 {    public static void
main(String[] args) {

    Scanner sc = new Scanner(System.in);

    // Input

    System.out.print("Enter an Integer value: ");
int intValue = sc.nextInt();

    System.out.print("Enter a Float value: ");
float floatValue = sc.nextFloat();

    System.out.print("Enter a Double value: ");
double doubleValue = sc.nextDouble();

    // Display Original Values

    System.out.println("\n----- Original Values -----");

    System.out.println("Integer : " + intValue);

    System.out.println("Float  : " + floatValue);

    System.out.println("Double : " + doubleValue);

    // Implicit Type Casting    double
implicitCast = intValue;    // Explicit
Type Casting    int explicitCast = (int)
doubleValue;

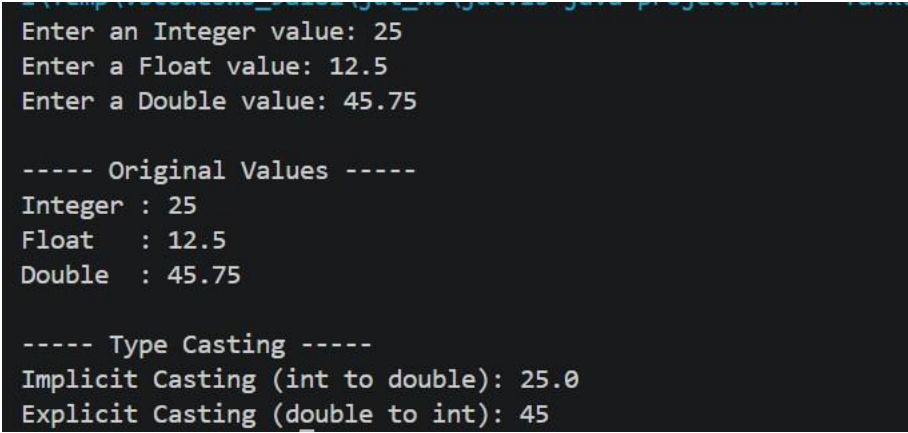
    // Display Type Casting

    System.out.println("\n----- Type Casting -----");
```

```
        System.out.println("Implicit Casting (int to double): " + implicitCast);
    System.out.println("Explicit Casting (double to int): " + explicitCast);

    sc.close();
}
}
```

Output:



```
Enter an Integer value: 25
Enter a Float value: 12.5
Enter a Double value: 45.75

----- Original Values -----
Integer : 25
Float   : 12.5
Double  : 45.75

----- Type Casting -----
Implicit Casting (int to double): 25.0
Explicit Casting (double to int): 45
```

Explanation:

This program accepts integer, float, and double values from the user and demonstrates both implicit and explicit type casting. It displays the values before and after conversion, making it easy to understand the changes that occur during type casting. This task improves knowledge of data conversion techniques used in Java programming.

Conclusion:

This Week 1 assignment helped me understand the fundamental concepts of Java programming. I learned how to write and execute Java programs, use variables and different data types, accept user input using the Scanner class, perform arithmetic operations using operators, and apply both implicit and explicit type casting. By completing these tasks, I improved my programming skills and gained confidence in writing simple Java programs. This assignment has provided a strong foundation for learning advanced Java concepts in the upcoming weeks.